SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3
Comparative Analysis

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks: 25**

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is **considering AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

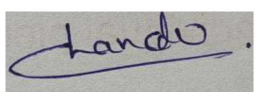
Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Code of Conduct:

I M. Phanith Chandu (192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

ANSWERS

Q1. CNN vs Fully Connected Neural Network

Comparative Analysis Table

Parameter	Fully Connected Neural Network	Convolutional Neural Network (CNN)
Architectural Design	Flat, fully-connected layers; every neuron connects to every neuron in the next layer	Layered design with convolution, pooling, and FC layers arranged hierarchically
Motivation	Designed for generic, non-spatial data (tabular inputs)	Designed specifically to exploit spatial structure present in images
Spatial Feature Extraction	Image must be flattened into a 1-D vector, destroying spatial relationships between pixels	Convolutional filters slide over the 2-D image, preserving spatial relationships between neighbouring pixels
Number of Parameters	Extremely high — a 256x256 X-ray image flattened gives 65,536 inputs, each connected to every hidden neuron	Much lower — a small filter (e.g., 3x3) is shared across the whole image, drastically cutting parameter count
Computational Complexity	High memory and compute cost; scales poorly with image resolution	Efficient — local receptive fields and shared weights reduce FLOPs and memory usage
Ability to Handle Image Data	Poor — loses edges, textures, and shape cues; prone to overfitting on high-dimensional pixel data	Strong — learns hierarchical features (edges → textures → organ structures) directly suited to diagnostic imaging

Justification / Analysis

- Convolutional layers apply small learnable filters across local regions of the X-ray image, detecting patterns such as edges, bone boundaries, and tissue density variations while preserving spatial context.
- Filters act as feature detectors; early filters capture low-level cues (edges, contrast) while later filters combine them into disease-relevant patterns (nodules, fractures, opacities).
- Pooling layers (max/average) down-sample feature maps, reducing dimensionality and computational load while making the learned features robust to small translations and noise in the scan.
- Because parameters are shared across spatial locations, CNNs require far fewer weights than a fully connected network of comparable depth, lowering overfitting risk on limited medical-imaging datasets.
- Given these advantages — reduced parameters, preserved spatial context, and translation robustness — a CNN is clearly the more suitable architecture for X-ray classification compared to a plain fully connected network.

Q2. Different CNN Layers and Filters

Comparative Analysis Table

Layer / Filter Type	Role in Defect Detection	Key Characteristics
Convolutional Layer	Extracts local visual features such as edges, cracks, scratches, and texture irregularities on the product surface	Uses learnable filters (kernels); preserves spatial information; low parameter count due to weight sharing
Pooling Layer	Reduces the spatial size of feature maps while retaining the most important defect signals	Down-samples via max/average pooling; adds translation invariance; lowers computation
Activation Layer (e.g., ReLU)	Introduces non-linearity so the network can model complex defect patterns beyond simple linear boundaries	Applied element-wise; computationally cheap; mitigates vanishing gradients compared to sigmoid/tanh
Fully Connected Layer	Combines all extracted high-level features to make the final defect/no-defect classification decision	Dense connections; high parameter count; typically placed at the end of the network
Early-Layer Filters	Detect low-level, generic features: edges, corners, colour gradients	Small receptive field; broadly reusable across different defect types
Deeper-Layer Filters	Detect high-level, defect-specific features: shape of a crack, pattern of a dent, texture anomaly	Large receptive field; combine low-level features into semantically meaningful defect signatures

Justification / Analysis

- Convolutional layers scan the product image locally, capturing fine-grained surface irregularities without losing their spatial location — essential for pinpointing where a defect occurs.
- Pooling layers compress this information, keeping the network efficient enough for real-time inspection on a production line while still retaining the strongest defect indicators.
- Activation layers allow the network to learn non-linear decision boundaries, letting it distinguish subtle real defects from harmless surface variations (lighting, reflections).
- Fully connected layers integrate the pooled, activated features into a single defect classification score, acting as the decision-making stage of the pipeline.
- Early filters build a generic vocabulary of edges and textures, while deeper filters recombine that vocabulary into specific, learned defect signatures — together this hierarchy lets the system move from raw pixels to an accurate, actionable defect classification.

Q3. Parameter Sharing vs Independent Weights

Comparative Analysis Table

Parameter	Independent Weights (Fully Connected Network)	Parameter Sharing (CNN)
Number of Parameters	Very high — a separate weight for every input-neuron connection	Very low — the same small filter (e.g., 5x5) is reused across the entire image
Memory Requirements	Large memory footprint; impractical for thousands of high-resolution frames	Compact memory footprint; feasible for continuous, high-volume video/image streams
Computational Complexity	Grows steeply with image resolution and network depth	Grows much more slowly, since the same filter weights are re-applied via sliding convolution
Feature Detection	Learns a unique, position-specific detector for each location — cannot generalize a pattern seen in one region to another	Learns one generic detector (e.g., edge/person-shape detector) that applies everywhere in the frame
Translation-Related Invariance	None — the network reacts differently if the same object appears in a different location in the frame	Built-in — the same filter fires consistently regardless of where the object appears, giving translation invariance

Justification / Analysis

- By reusing the same filter weights at every spatial location, CNNs cut the parameter count by orders of magnitude compared to a fully connected network processing the same high-resolution frames.
- This directly reduces memory and compute requirements, allowing the surveillance system to process thousands of frames per hour on affordable hardware.
- Because a single filter learns a general pattern (e.g., a person's silhouette or a moving vehicle edge), it can detect that pattern anywhere in the frame — this is the source of translation invariance.
- Independent weights, by contrast, force the network to relearn the same pattern separately for every pixel location, wasting capacity and requiring far more training data to generalize.
- For large-scale, real-time image processing such as surveillance, parameter sharing is essential — it makes the system computationally feasible while improving generalization across camera views and object positions.

Q4. AlexNet vs ResNet

Comparative Analysis Table

Parameter	AlexNet	ResNet
Network Depth	Shallow — 8 layers (5 convolutional + 3 fully connected)	Very deep — variants from 18 to 152+ layers
Layer Organization	Sequential stack of conv, pooling, and FC layers	Organized into residual blocks with skip/shortcut connections
Convolutional Filters	Large filters in early layers (11x11, 5x5)	Small, uniform filters (mostly 3x3) stacked in deep residual blocks
Parameter Requirements	~60 million parameters, concentrated in FC layers	Fewer parameters relative to depth due to smaller filters and global average pooling instead of large FC layers
Feature Extraction Capability	Good for moderately complex tasks; limited depth restricts hierarchical abstraction	Excellent — extreme depth captures highly abstract, hierarchical features
Training Difficulty	Relatively easier to train given shallower depth	Would be very hard to train at such depth without residual connections
Vanishing-Gradient Problem	Present and worsens if depth is increased naively	Solved via identity skip connections that let gradients flow directly through the network
Computational Cost	Lower total depth but heavy FC layers increase cost there	Higher raw depth, but efficient design keeps overall cost manageable relative to accuracy gained
Classification Performance	Strong for its era (ILSVRC 2012 winner) but surpassed by deeper models	State-of-the-art accuracy on large, complex, fine-grained classification tasks

Justification / Analysis

- AlexNet's shallow depth limits how many levels of abstraction it can learn, making it less suited to complex recognition tasks with many fine-grained classes.
- ResNet's residual (skip) connections allow the network to be trained much deeper without suffering from vanishing gradients, since gradients can flow directly through identity shortcuts during backpropagation.
- This lets ResNet learn richer, more hierarchical feature representations, which translates into higher accuracy on complex, large-scale image recognition benchmarks.

Q5. Regularized CNN vs Non-Regularized CNN

Comparative Analysis Table

Parameter	CNN without Regularization	CNN with Dropout, Data Augmentation & Batch Normalization
Overfitting	High — model memorizes training leaf images, including noise and background artifacts	Significantly reduced — dropout and augmentation force the network to learn generalizable disease features
Generalization	Poor performance on unseen leaf images from new plants/conditions	Strong performance on unseen images due to exposure to varied, augmented samples
Training Stability	Can be unstable; internal covariate shift slows and destabilizes convergence	More stable — batch normalization normalizes layer inputs, smoothing the loss landscape
Computational Requirements	Lower per-epoch cost but often needs more epochs / careful tuning to avoid overfitting	Slightly higher per-epoch cost, but converges more reliably and reduces wasted training cycles
Model Performance	High training accuracy, low real-world/deployment accuracy	Balanced training and validation accuracy; better real-world reliability

Justification / Analysis

- Dropout randomly deactivates neurons during training, preventing the network from relying too heavily on any single feature and reducing memorization of training-specific leaf artifacts.
- Data augmentation (rotation, flipping, brightness/contrast changes) simulates the natural variation seen in real field images — different lighting, leaf orientation, and camera angles — improving robustness.
- Batch normalization stabilizes and speeds up training by normalizing intermediate activations, which also has a mild regularizing effect that further curbs overfitting.
- Given the initial model's high training accuracy but poor generalization, applying all three techniques together — dropout, data augmentation, and batch normalization — is the most appropriate approach for a reliable, field-deployable plant disease classifier.